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RE...	remove mc3...	9 months ago

Getting Started Guide: Microchip PIC-IoT WA (Wireless for Amazon Web Services) Application

Devices: |
PIC24FJ128GA705(MCU) |
WINC1510(Wi-Fi®) |
ECC608(CryptoAuthLib) |

About

No description or website provided.

#iot #cloud #pic24 #ecc608 #winc1510

Readme
8 stars
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10 forks

Releases 4

4.1.3 Latest
on 2 Dec 2021

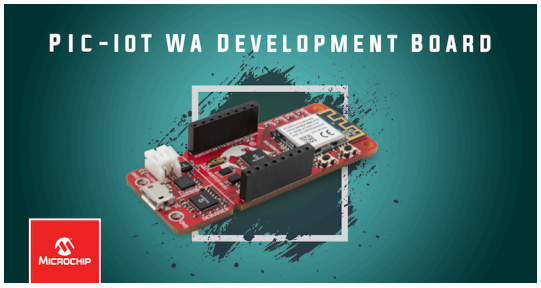
+ 3 releases

Packages

No packages published

Languages

- C 87.2%
- Makefile 12.7%
- Shell 0.1%



Getting Started Guide

- Below describes the Out of Box (OOB) operation of the development board.
- More detailed information can be found further below this document.
- For accessing production hex files, release notes, and Known Issues please click the release tab

Materials

- Internet Connection Device
- Wifi / Network Device
- Personal Computer
- Serial Terminal (optional)

Operation

1. Connect board to PC using USB-micro cable.
    - The LEDs will **Cycle** upon startup: **BLUE-->GREEN-->YELLOW-->RED**.
    - Short delay: **BLUE--**

>GREEN-->YELLOW-->RED.

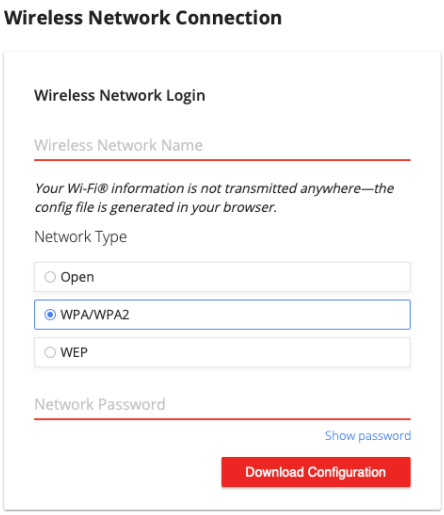
2. The **BLUE LED** will begin to blink, this indicates the board is attempting to join the local **ACCESS POINT**.

3. On connecting to **Access Point** the blinking will stop, and the **LED** will become **STATIC**.

    - By **Default**, the device will attempt to use:
    - **WPA/WPA2**
    - Network Name: **MCHP.IOT**
    - Password: **microchip**

4. To use custom Credentials, the board will appear on the PC enumerated as a mass storage device under the name **CURIOSITY**.

    - Credentials can be downloaded as the file **WIFI.CFG** using the **CLICK-ME.HTM** file stored on the **CURIOSITY** device.

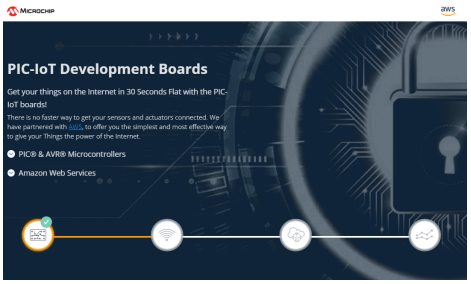


5. After becoming connected to the **ACCESS POINT**, the **GREEN LED** will begin to blink, this indicates the board is attempting to establish a web connection with cloud providing service.
    - **GREEN LED** will stop blinking and remain **ON** when connection is established.
    - Using the in-module TCP/IP stack pre-configured with provisioned credentials; the device establishes a **MQTT** connection with the IoT Broker provider (AWS).

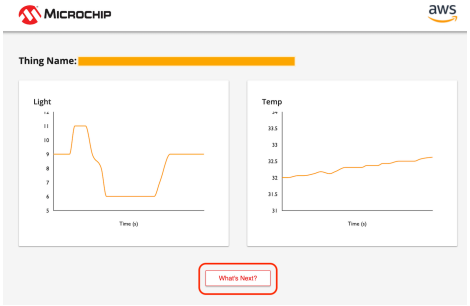
6. After successfully establishing MQTT connection, the **YELLOW LED** will blink. (250 mSec)

- Indicating data exchanged between the End-Device (PIC-IoT), and BROKER (AWS). (every (1) Sec)

7. Connect to the www.pic-iot.com/pic-iot/aws/**{thingName}** website to view publish/subscribe data.
   - **{thingName}** is the unique identifier for the development board.
   - This page can be also be found via launching the **CLICK-ME.HTM** file on the **CURIOSITY** device.
   - This page can be also be found by scanning the QR code on the back of the development board.

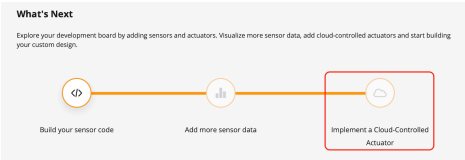


8. There will be (2) scrolling graphs visible.
   - (1) shows temperature sensor
   - (1) shows the light sensor value.

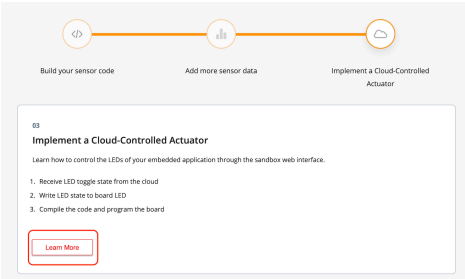


9. Click on the **What's Next** button beneath the graphs to peform action(s) from the cloud.

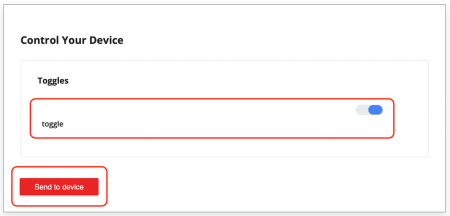
10. Select the **Implement a Cloud-Controlled Actuator** to demostrate cloud performed behaviors.



11. Click on the **Learn More** button to expand page interface. + Scroll to the bottom of **Step 5** where a panel will read **Control Your Device**.



12. By default only a **Toggle** feature is demostrated. + **Custome implmentations are described further on above the panel.**



13. Trigger an action by simply pressing the **Send to device** button beneath actuator fields. + All Actuator data is sent as one message when the **Send to device** button is pressed. + When **Toggled** the **YELLOW LED** will remain on for **(2) Seconds**. + When unselected, the **YELLOW LED** will remain off for **(2) Seconds**. + Because **Toggle** manipulates the **desired** state; the state **must** be changed to observe the behavior.

14. If desired, additional basic messaging can be seen to a connected Serial Terminal at the expected 9600 baud rate.

Requirements

- **MPLAB® X Integrated Development Environment (IDE) v5.25 or later**: MPLAB® X IDE is a computer software program based on the open source NetBeans IDE from Oracle. It is used to develop applications for Microchip microcontrollers and digital signal controllers. It runs on Windows®, Mac OS® and Linux®. For the latest version, please refer to: MPLAB-X

- **MPLAB® XC16 Compiler v1.41 or later**: MPLAB® XC compilers support all of Microchip's PIC, AVR and dsPIC devices where the code is written in the C programming language. XC16 is the recommended compiler for 16-bit PIC MCUs. In this lab, as well as with the succeeding labs, you will be using MPLAB® XC16 for an PIC MCU. For the latest version, please refer to: XC-Compiler

- **Compiler Optimization** Compilation of source code can be achieved using supporting MPLAB® XC compiler: **XC16**.

- XC16 Compiler v1.41 or later: Supported by optimization **level -O0, -O1 (free)** and **-**

O2, -Os (pro)

MPLAB® X IDE XC16 support all 16-bit PIC® microcontrollers (MCUs). This is a internally developed compiler which is specially designed to maximize features available to the PIC® microcontrollers (MCUs).

- **MPLAB® Code Configurator (MCC) Not Supported**: This PIC-IoT demonstration does not support reconfiguration of peripherals using MCC.
- **PIC IoT Development Board**: The PIC-IoT development board combines a powerful 16-bit PIC24FJ128GA705 MCU, an ATECC608A CryptoAuthentication™ secure element IC and the fully certified ATWINC1510 Wi-Fi® network controller - which provides the most simple and effective way to connect your embedded application to a cloud platform. The board also includes an on-board debugger and requires no external hardware to program and debug the MCU.

Application Scope

The PIC-IoT development board has been created with the intention of demostrating a one source solution for evaluation of existing cloud provider solutions. This example end-device leverages the catalog of devices, and libraries provided through Microchip's extensive product line to showcase a basic Internet of Things product connection. Data exchange between server and in field device is implemented using on board sensors for temperature, and light value observations. Behavior actions are demonstrated through visual indication of the 'Data' LED as controlled through the Web based APIs.

General Out-Of-Box operation is as described below:

1. Use the On-Board WINC1510 WiFi module to establish local WiFi connection to Router/Switch or Network source.
2. The WINC1510 through use of the ECC608 Cyrpo establishes a Secure (TLS) Socket Connection with select Cloud Provider using a TCP connection. The Blue 'WiFi' LED is used to indicate this status.
3. Using the Microchip Software

Library for the Message-Queuing-Telemetry-Transport (MQTT) standard format; data is exchanged between client (end-device) and broker (cloud). The Green 'Connect' LED is used to indicate this status.

4. Sensor Data is sent as Telemetry Data between device and broker at a near periodic rate of (1) Second. The Yellow 'Data' LED is used to indicate this status.
5. Capture of Data sent from Broker to Device can be observed through a Serial terminal when USB-Micro is connected to development board.
6. Behavior variation can be observed on the 'Data' LED when triggered through the web based API, and sent through the broker to end device using MQTT protocol transported through the TCP connection securely established using the ECC608 Crypto device.
7. The Push button has no effect outside of variation of start-up operation; refer to User Guide for additional information regarding Soft-AP operation.

- The Red 'Data' LED remaining on may indicate a hardware fault issue with the development board.

Application Description

Cloud Platforms

- AWS
  i. Publish payload for sensor data (telemetry)
     - topic: `thingName/sensors`
     - payload:

```
{
    "Light": lightV
    "Temp": tempera
}
```

  ii. Device publishes payload to update the Device Shadow
     - topic: `$aws/things/thingName/shadow/update`
     - payload:

```
{
    "state":
```

```
            {
                "reported"
                {
                    "togg
                }
            }
        }
```

iii. User Interface publishes payload to Device Shadow

- topic:

  `$aws/things/thingName/shadow/update`

- payload:

```
{
    "state":
    {
        "desired":
        {
            "toggle"
        }
    }
}
```

iv. Device subscribes to delta to receive actionable changes

- topic:

  `$aws/things/thingName/shadow/update/delta`

- payload:

```
{
      "state":
     {
          "Light": light
     }
}
```

The PIC IoT development board publishes data from the on-board light and temperature sensor every second to the cloud.

The data received over the subscribed topic is displayed on a serial terminal.

Sending MQTT publish packets

- The C code for sending MQTT publish packets is available in PICIoT.X/mcc_generated_files/application_manager.c file.
- The API `static void sendToCloud(void)` is responsible for publishing data at an interval of 1 second.

Sending MQTT subscribe packets

- The C code for sending MQTT subscribe packets is available in

PICIoT.X/mcc_generated_files/application_manager.c file.

- The API `static void subscribeToCloud(void)` is responsible for sending MQTT subscribe packets to the cloud after MQTT connection is established.

Processing Packets received over subscribed topic

- The C code for processing MQTT publish packets received over the subscribed topic is available in PICIoT.X/mcc_generated_files/application_manager.c file.
- The `static void receivedFromCloud(uint8_t *topic, uint8_t *payload)` function is used for processing packets published over the subscribed topic.

Secure Provisioning & Transport Layer Security

1. The PIC-IoT board under the Wireless for Amazon Web Services (WA) variation is shipped pre-provisioned for coordination with the AWS Cloud system.

2. Security is achieved by using the built-in Transport Layer Security (TLS) 'stack' configured within the WINC Wi-Fi Module.
3. A Pre-Manufacturing process has configured the appropriate Slot locations on the ATECC608A security device.
4. All required certificates used for signing and authentication have been written to, and 'locked' into allocated slots.
    - This process is achieved through: TrustFlex
    - Additional options are also supported, such as: Trust&Go
    - Fully scope of support for Secure Aspects of development can be found here: Trust Platform
5. This process has been performed to allow for an Out Of Box (OOB) operation of the PIC-IoT development board along with supporting web page.
6. Shadow Topic are a key application feature being supported through the AWS platform. Further reference can be found here:
    - Shadow MQTT Topics
7. For PIC-IoT development not

provisioned for the AWS platform; refer to the below repo location:

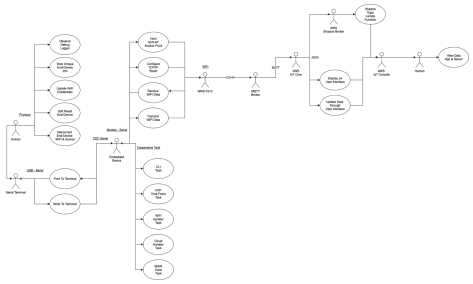
- **gitHub Provisioning Repo coming in future. Currently refer to:**
- AVR-IoT AWS Provisioning

Understanding the Device Shadow in AWS

README.md

The Shadow Topics are used to retain a specific value within the Broker, so End-Device status updates can be managed.

- Shadow Topics are used to restore the state of variables, or applications.
- Shadow Topics retain expected values, and report if Published data reflects a difference in value.
- When difference exist, status of the delta is reported to those subscribed to appropriate topic messages.



2. Updates to the device shadow are published on $aws/things/{ThingName}/shadow/update topic. When a message is sent to the board by changing the value of the **toggle** fiels in **Control Your Device** section:

    - This message is published on the $aws/things/{ThingName}/shadow/update topic.
    - If the curent value of **toggle** in the device shadow is different from the **toggle** value present in the AWS Device Shadow, the AWS Shadow service reports this change to the device by publishing a message on $aws/things/{ThingName}/shadow/update/delta topic.
    - The JSON structure of the message sent should appear as below

    ```
    {
      "state": {
    ```

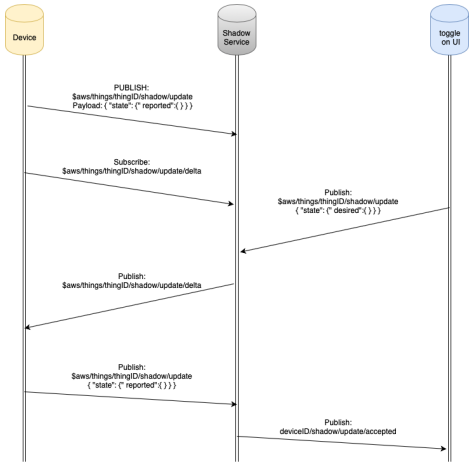
```
            "desired": {
              "toggle": [va
            }
          }
        }
```

3. The meta data, and delta difference will be reported via the Serial Terminal upon a difference between desired/reported.

4. In response to this, the end device publishes a message to $aws/things/{ThingName}/shadow/update topic.
   - This message is published to **report** the update to the **toggle** attribute.
   - The JSON structure of the message sent should appear as below

```
{
  "state": {
    "reported": {
      "toggle": [value]
    }
  }
}
```

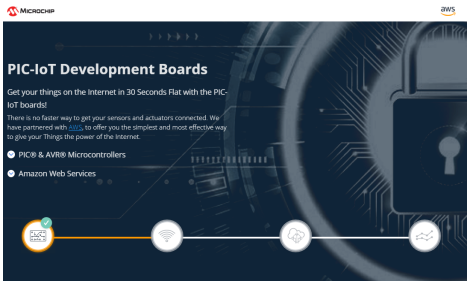
5. Application flow when using the device shadow



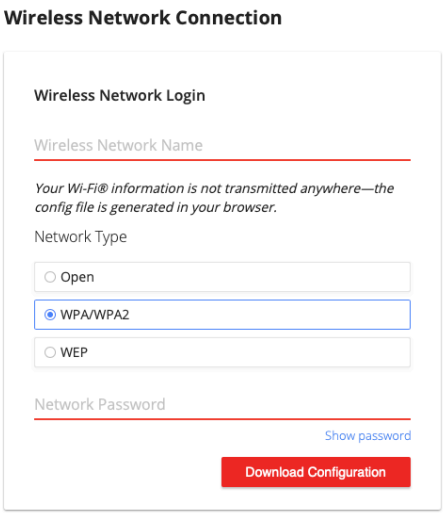
Detailed Operation

0. There are three possible variations within application behavior possible by holding push buttons on startup
    - Default behavior: No Button Pressed
    - Soft AP: SW0 is Held on startup (see description farther in document)
    - Default behavior Restore DEFAULT Credentials: SW0 & SW1 Held on startup. This state is reflected by **BLINKING GREEN LED** until a Wi-Fi connection is established.
        - After a successful connection; last VALID CREDIENTIALS are maintained in the WINC for next power cycle connection.

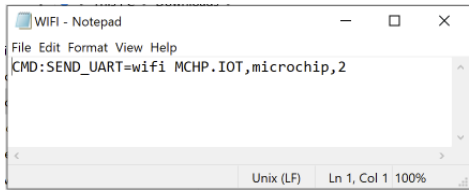
1. Connect board to PC using USB-micro cable.
    - The LEDs will **Cycle** upon startup: **BLUE-->GREEN-->YELLOW-->RED**, short delay, **BLUE-->GREEN-->YELLOW-->RED**.
2. The **BLUE LED** will begin to blink, this indicates the board is attempting to join the local **ACCESS POINT**.



3. Update the Wi-Fi Credentials; upon connecting the blinking will stop, and the **LED** will become **STATIC**. Below are the easiest methods to update credentials.
    - The board will appear on the PC enumerated as a mass storage device under the name **CURIOSITY**. Credentials can be downloaded as the file **WIFI.CFG** using the **CLICK-ME.HTM** file stored on the **CURIOSITY** device.



- This will launch the URL: https://pic-iot.com/pic-iot/aws/{ThingName}.
- After entering credentials, the **.CFG** file is produced through the web browser. No information is shared through the internet.
- Drag and Drop, or Copy and Paste the **WIFI.CFG** file onto the **CURIOSITY** device to load new credentials onto the IoT demonstration board.

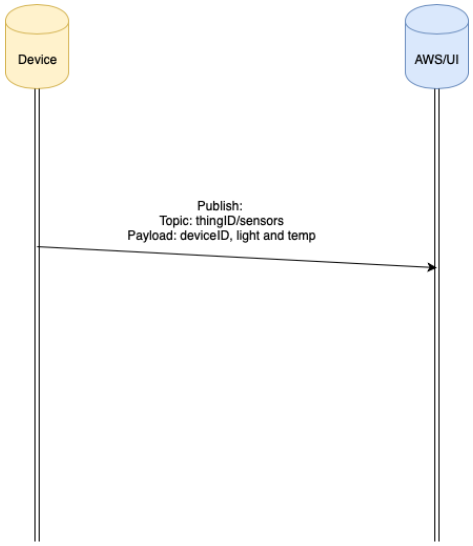


- Use a **Serial Terminal** to update the WiFi Credentials loaded onto the WINC module. Use the Command Line Interface (CLI) supported

command `wifi host_name,pass_code,auth_type` | host_name/pass_code are entered strings, auth_type is int value: (0: open, 1: WEP, 2: WPA).

4. After becoming connected to the ACCESS POINT, the GREEN LED will begin to blink, this indicates the board is attempting to establish a TCP/IP and MQTT connection with the cloud providing service. The GREEN LED will stop blinking and become STATIC when the TCP and MQTT connection is established.
    - Using the in module TCP/IP stack pre-configured with provisioned credentials; the device establishes a **MQTT** connection with the IoT Broker provider (AWS).

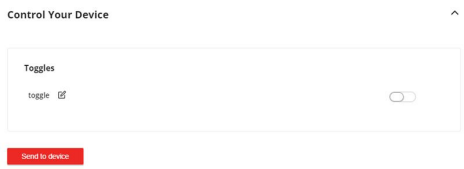
5. After successfully establishing MQTT connection, the **YELLOW LED** will blink, indicating data exchanged between the End-Device (PIC-IoT), and BROKER (AWS).



6. Connect to the www.avr-iot.com/aws/{thingName}, or www.pic-iot.com/aws/{thingName}, device specific website to view publish/subscribe data.
    - This page can be found via launching the **CLICK-ME.HTM** file on the **CURIOSITY** device.
    - There will be (2) scrolling graphs visible. (1) shows temperature sensor, (1) shows the light sensor value.
    - Additional graphs can be produced altered through the published

topic message.

7. **Control Your Device** using the (3) rows beneath the '**Control Your Device**' section used to publish subscription data to end-devices through the broker.
    - **Only the use of Toggle is supported natively**
    - **Expanding features would require custom written Firmware implementation**
    - These example rows demonstrate options for: Toggle (boolean), Text Field (String), Sliders (integer)



8. When connection is established with the Broker, the publish message topic will be printed to a serial terminal through the CDC-USB bridge.
    - 9600 is expected Baud

Rate.

- When a topic subscription is received, the payload is printed in JSON format to the terminal.
- Topic subscription message are sent when the 'Send to device' push button on the webpage is pressed.

9. When the 'Desired' state is updated in the 'Delta' Shadow Topic.

   - Device which required updates to the 'Desired' state which differs from their last 'Reported' value will receieve a published message on the '.../delta' MQTT topic.
   - Upon reception the device will report the updated 'Desired' value for the attribute with timestamp on the console.

Additional Operations

Command Line Interface

- The PIC-IoT development board can also be accessed through a serial command line interface.
- This interface can be used to provide diagnostic information.
- To access this interface, use any preferred serial terminal application (i.e. Teraterm, Coolterm, PuTTy) and open the serial port labeled Curiosity Virtual COM port, with the following settings:

-	-
Baud Rate	9600
Data	8-bit
Parity Bit	None
Stop Bit	1 bit
Flow Control	None
Additional Settings	Local Echo: On
Transmit to the Microcontroller	CR+LF (Carriage Return + Line Feed)

Note: For users of the Windows environment, the USB serial interface requires the installation of an USB serial port driver.

Command	Arguments	D
reset	-	I s t
device	-	un
thing	-	Pr tl
reconnect	-	Re co t
version	-	ve s us
cli_version	-	co ve s us
wifi	Network SSID, Password, Security Option	I F aut
		P

debug	Debug Options	m se

wifi Security Options:

- 0 : Open - Password and Security option parameters are not required.
- 1 : WPA/WPA2 - Security Option Parameter not required.
- 2: WEP - Network Name, Password, and Security Option (3) Parameter are required when connecting to a WEP network. For example, 'wifi MCHP.IOT,microchip,3'.

debug Debug Options: Type in a number from 0 to 4; for the number of debug messages with 0 - the result is printing no messages and with 4 for printing all the messages.

- 0 : Minimal
- 1 : Critical
- 2 : Warning
- 3 : Info
- 4 : All

Soft Access Point (AP)

The PIC-IoT development board can be accessed through a Wi-Fi access point enabled by the Software-Enabled Access mode of the WINC1510. This can be another way to connect the board to a Wi-Fi network. To enter Soft AP mode, press and hold the SW0 push button before plugging in the board. When connecting to the module hosted access point, the user will need to enter the desired SSID and password credentials for the network. After the user enters the details, pressing the Connect button will reconfigure network credentials for the device.

AWS Multi Account Registration

This feature is in Beta phase. The user can leverage upon the provisioning tool to register existing PIC-IoT board in a personal account. More information on using the provisioning tool can be found in README.txt and iotprovsion.md which are part of the tool.

By default the AWS endpoint URL is read from WINC's memory. Upon using the provisioning tool the Microchip sandbox AWS endpoint URL present in WINC will be replaced by the personal account AWS endpoint URL. In those boards which do not carry the AWS endpoint URL in WINC's memory the Microchip Sandbox AWS endpoint URL provided in the macro `AWS_MCHP_SANDBOX_URL` will be used.

The firmware supports the use of custom AWS endpoint URL. This feature can be enabled/disabled through the macro `USE_CUSTOM_ENDPOINT_URL` in "cloud_config.h". Enabling this feature will ensure the AWS endpoint defined by the macro `CFG_MQTT_HOSTURL` in "mqtt_config.h" is used instead of the one present in WINC's memory for establishing the connection with the cloud. To switch to the usage of default AWS endpoint present in WINC's memory, disable this custom AWS endpoint URL by setting the macro `USE_CUSTOM_ENDPOINT_URL` to '0'.

In either case of using the default or custom option,the AWS endpoint URL being used can be observed on the serial terminal right before the board's start up LED cycle. Make sure to have the `ENABLE_DEBUG_IOT_APP_MSGS` enabled to get the IoT application layer specific logs printed out on the terminal,

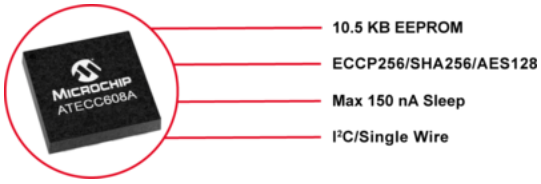
Software Features

WINC

Microchip's WINC1510 is a low-power consumption 802.11 b/g/n IoT (Internet of Things) module, specifically optimized for low-power IoT applications. The module integrates the following: Power Amplifier (PA), Low-Noise Amplifier (LNA), switch, power management, and a printed antenna or a micro co-ax (u.FL) connector for an external antenna, resulting in a small form factor (21.7 x 14.7 x 2.1 mm) design. It is interoperable with various vendors' 802.11 b/g/n access points. This module provides SPI ports to interface with a host controller. The WINC1510 provides internal Flash memory as well as multiple peripheral interfaces, including UART and SPI. The only external clock source needed for the WINC1510 is the built-in, high-speed crystal or oscillator (26 MHz). The WINC1510 is available in a QFN package or as a certified module.

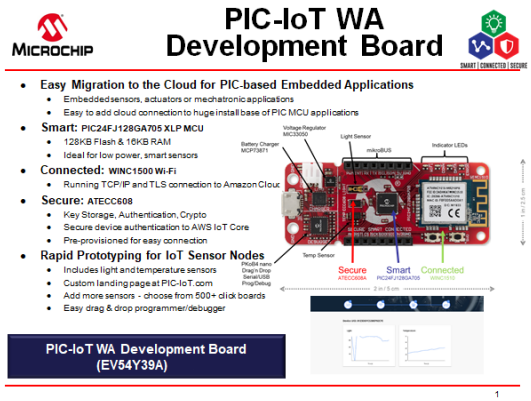
Crypto Authentication Library (CAL)

The ATECC608A is a secure element from the Microchip CryptoAuthentication™ portfolio with advanced Elliptic Curve Cryptography (ECC) capabilities. With ECDH and ECDSA being built right in, this device is ideal for the rapidly growing IoT market, by easily supplying the full range of security such as confidentiality, data integrity, and authentication to systems with MCUs or MPUs running encryption/ decryption algorithms. Similar to all Microchip CryptoAuthentication products, the new ATECC608A employs ultra-secure, hardware-based cryptographic key storage and cryptographic countermeasures, which eliminates any potential backdoors linked to software weaknesses.



Message Queuing Telemetry Transport (MQTT)

MQTT is a lightweight M2M (Machine to Machine) messaging protocol used to exchange data between devices. This is an established protocol, additional references can found through MQTT.org

The version used with the demostration is a generated software library produced through the MPLAB Code Configurator (MCC) tool, and has been written by Microchip for use family of devices.

Hardware Description